INCH-POUND

MIL-DTL-7793/1B
18 March 2005
SUPERSEDING
MIL-M-7793/1A
15 September 1988

DETAIL SPECIFICATION SHEET

METER, TIME TOTALIZING, MINIATURE, DIGITAL, 28 V DC, SQUARE

Inactive for new design after 14 April 1999.

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the meter described herein shall consist of this specification sheet and MIL-M-7793.

REQUIREMENTS:

1. Configuration. The meter shall meet the requirements on figure 1 and in table I.

2. Tolerances. Unless otherwise specified in MIL-M-7793 or this specification sheet, tolerances are ±0.02 inch for two-place decimals and ±0.015 inch for three-place decimals.

3. Mounting flange. Meters with fixed flanges shall be tin plated on the front surface of the mounting flange in accordance with Class C of ASTM-B545, " Tin, Electrodeposited Coatings of," or ASTM-B339, "Pig Tin," to provide an electrical ground for the case (see NOTE 2 of this specification sheet).

4. Life accuracy. Meter accuracy shall be ±0.1 percent from -85 to 257 °F (-65 to 125 ºC) and across the maximum and minimum input operating voltage range specified in table I.

5. Power consumption. Power consumption at 28 volts direct current (V dc) shall be not greater than 85 milliwatts.

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6. Ripple voltage. The meter shall continue to operate as specified in MIL-M-7793 when subjected to a cyclic peak of ripple voltage (see NOTE 3) of less than 2.0 V dc and the frequency-voltage coordinates on figure 2.

6.1 Ripple voltage measurement. The ripple voltage shall be measured with a peak reading voltmeter in series with a 4.0-microfarad capacitor. The higher of the two values measured when the voltmeter is successively connected to each of two polarities shall be considered the ripple voltage.

6.2 Frequency characteristics. The frequency components of the ripple voltage shall be within the limits on figure 2 when measured as conducted interference.

7. Operating temperature range. The operating temperature range shall be from -85 to 257 °F (-65 to 125 ºC).

8. Time range. Total readout of the meters shall be 9,999 hours.

9. Line drop. Line drop (see NOTE 4) shall be less than 2 V dc.

10. Transient protection. No temporary or permanent degradation shall be produced in the meter for the input voltage and time values shown on figure 3 and figure 4 (80 V and 600 V transient, respectively).

10.1 Transient voltage. Transient surge voltages shall be within the limits on figure 3 when converted to their equivalent step functions. Transient spike voltages shall be within the limits shown on figure 4. They shall be not greater than ±600 V nor depart from the normal wave form for more than 50 microseconds.

11. Dielectric test voltages. The dielectric test voltages shall be 350 V root mean square (rms) at altitude specified in MIL-M-7793 and 600 V rms at room ambient pressure.

12. Random vibration. Vibration testing shall be performed in accordance with MIL-M-7793.

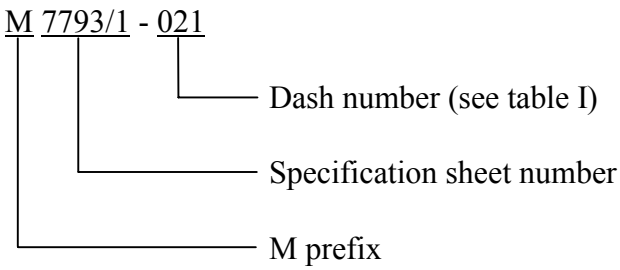
13. Qualification tests. Qualification tests shall be conducted in accordance with MIL-M-7793, except that the accuracy, voltage, temperature, and dielectric test voltages shall be as specified herein.

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NOTES:

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

1. Part or identifying number (PIN). The PIN to be used for meters acquired to this specification sheet is created as follows:



2. Meter flanges. Meters may be furnished with no flange, flange with various mounting depths, or with adjustable split mount flange in accordance with figure 1 (see MS3345-2).

3. Ripple voltage. The cyclic variation of voltage about the mean level of the dc voltage during steady-state dc electric system operation.

4. Line drop. The voltage difference between the point of voltage regulation and the power input terminals of the meter.

5. Supersession data. The meters covered by dash numbers shown on MS21341 were canceled after 15 September 1994. Use the dash numbers given in table I. Use only the MIL-DTL-7793/1 meters for substitution. Supersession information for MS21341A and MS21341B is given in tables II and III, respectively.

6. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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TABLE I. Meter length, weight, and input operating voltage.

Dash no. 1/	A Flange data (inches)	B Case length (inches)		Maximum weight 2/ (ounces)	Input operating voltage	
		Maximum	Minimum		Maximum	Minimum
001	Adjustable per MS3345-2	1.094	1.047	0.6	34	10
002	No flange					
003	0.000 (flush)					
004	0.031					
005	0.062					
006	0.094					
007	0.125					
008	0.156					
009	0.188					
010	0.218					
011	0.250					
012	0.281					
013	0.312					
014	0.344					
015	0.375					
016	0.406					
017	0.438					
018	0.469					
019	0.500					
020	0.531					
021	0.562					
022	0.594					
023	0.625					
024	0.656					
025	0.688					
026	0.719					
027	0.750					
028	0.781					

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TABLE I. Meter length, weight, and input operating voltage - Continued.

Dash no. 1/	A Flange data (inches)	B Case length (inches)		Maximum weight, 2/ (ounces)	Input operating voltage	
		Maximum	Minimum		Maximum	Minimum
029	0.000 (flush)	2.062	2.000	1.5	32	23
030	0.031					
031	0.062					
032	0.094					
033	0.125					
034	0.156					
035	0.188					
036	0.218					
037	0.250					
038	0.281					
039	0.312					
040	0.344					
041	0.375					
042	0.406					
043	0.438					
044	0.469					
045	0.500					
046	0.531					
047	0.562					
048	0.594					
049	0.625					
050	0.656					
051	0.688					
052	0.719					
053	0.750					
054	0.781					

1/ Dash numbers M7793/1-001 through -028 are replacements for MS21341A meters and dash numbers M7793/1-029 through -054 are replacements for MS21341B meters. See tables II and III.

2/ Maximum weight is measured with flange.

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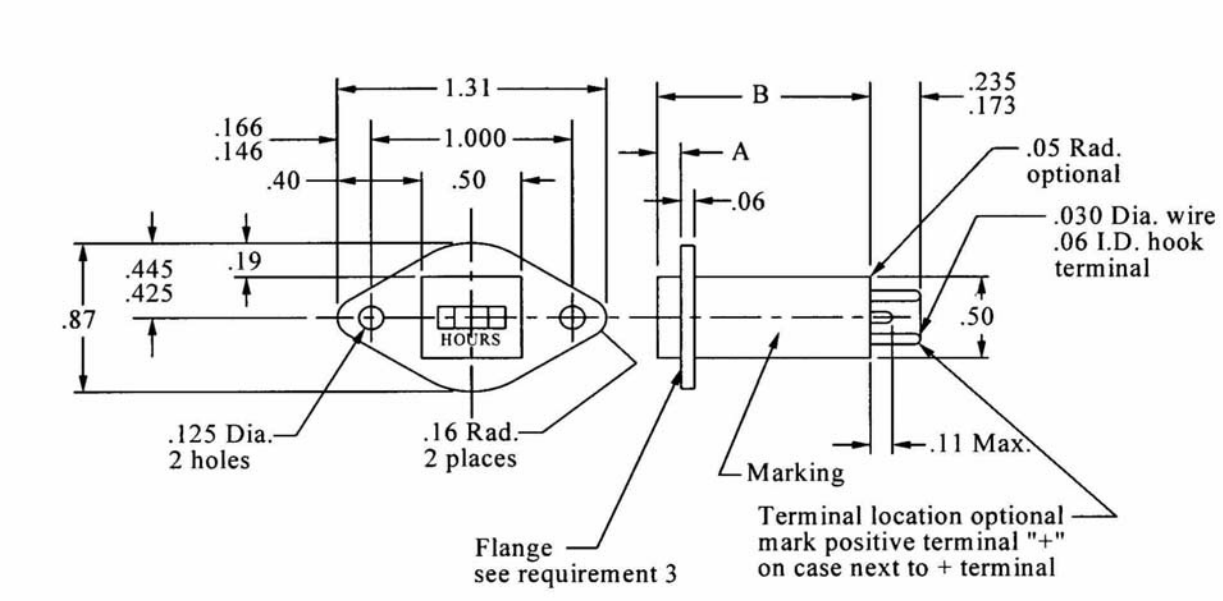
TABLE II. MS21341A substitution table.

Canceled meter MS21341A dash numbers	Substitute meter M7793/1 dash numbers	Canceled meter MS21341A dash numbers	Substitute meter M7793/1 dash numbers
01	003	14	016
02	004	15	017
03	005	16	018
04	006	17	019
05	007	18	020
06	008	19	021
07	009	20	022
08	010	21	023
09	011	22	024
10	012	23	025
11	013	24	026
12	014	25	027
13	015	26	028

TABLE III. MS21341B substitution table.

Canceled meter MS21341B dash numbers	Substitute meter M7793/1 dash numbers	Canceled meter MS21341B dash numbers	Substitute meter M7793/1 dash numbers
01	029	14	042
02	030	15	043
03	031	16	044
04	032	17	045
05	033	18	046
06	034	19	047
07	035	20	048
08	036	21	049
09	037	22	050
10	038	23	051
11	039	24	052
12	040	25	053
13	041	26	054

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Notes:

1. All dimensions are in inches.
2. For dimensions A and B see table I.

FIGURE 1. Meter configuration.

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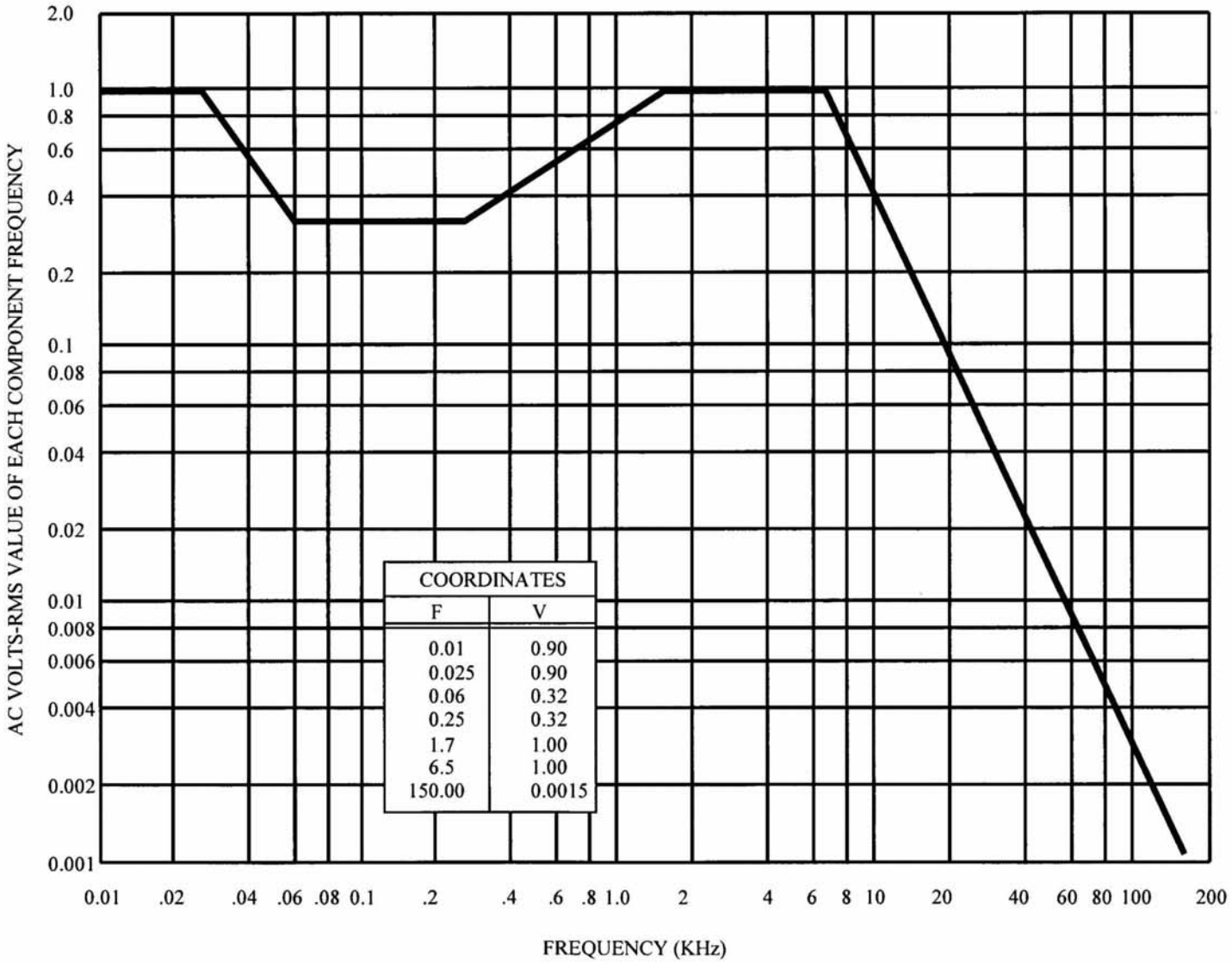


FIGURE 2. Frequency characteristics of ripple in 28 V dc electric systems.

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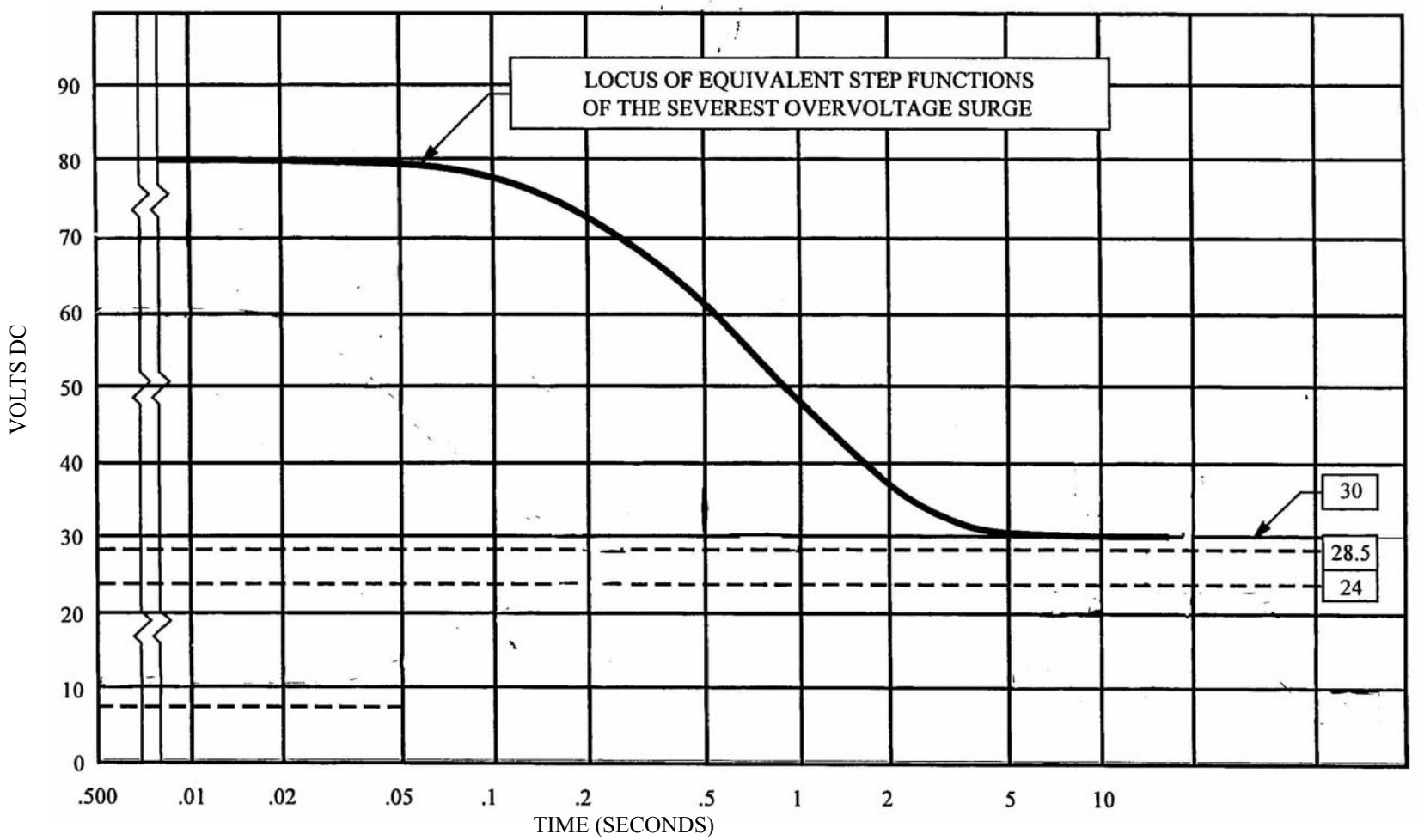


FIGURE 3. Transient surge step function loci limits.

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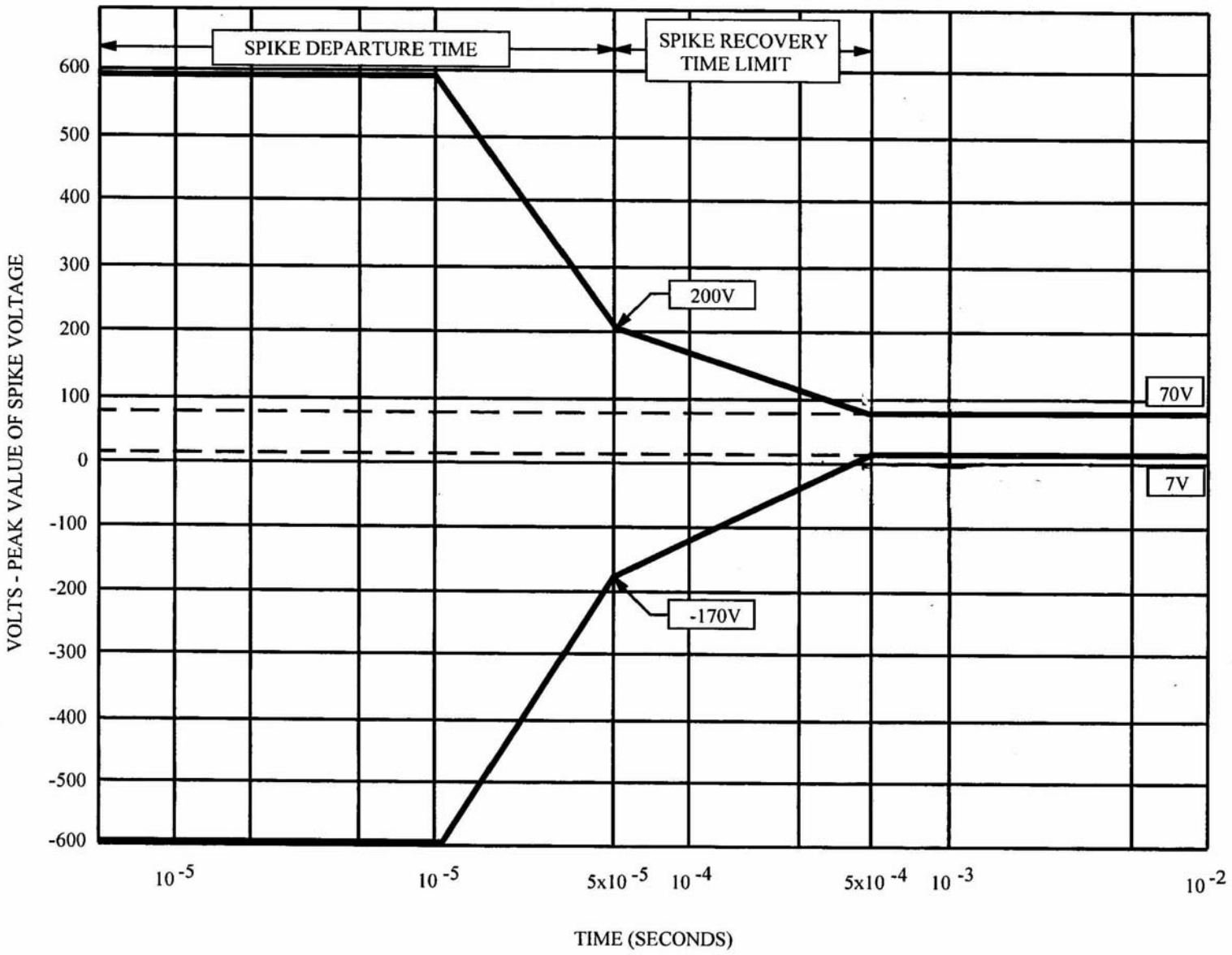


FIGURE 4. Envelope of spike voltages for direct current equipment.

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